IMO(33^{rd}) – 1992

First Day

1. Find all integer triples (p, q, r) such that $1 < p < q < r$ and $(p-1)(q-1)(r-1)$ is a divisor of $(pqr-1)$.
2. Find all functions $f : R \to R$ such that $f(x^2 + f(y)) = y + f(x)^2$, for all $x, y \in R$.
3. Given nine points in space, no four of which are coplanar, find the minimal natural number n such that for any coloring with red or blue of n edges drawn between these nine points there always exists a triangle having all edges of the same color.

Second Day

4. In a plane, let there be given circle C, a line l tangent to C, and a point M on l. Find the locus of points P that has the following property: There exist two points Q and R on l such tha M is the midpoint of QR and C is the incenter of PQR.
5. Let V be a finite subset of Euclidean space consisting of points (x, y, z) with integer coordinates. Let S_1, S_2, S_3 be the projection of V onto yz, zx, xy planes, respectively. Prove that $|V|^2 \le |S_1||S_2||S_3|$ where $|X|$ denotes the number of elements of X.
6. For each positive integers n, denote $s(n)$ the greatest integer such that for all positive integer $k \le s(n)$, n^2 can be expressed as a sum of squares of k positive integers:
   (a) Prove that $s(n) \le n^2 - 14$, for all $n \ge 14$.
   (b) Find a number n such that $s(n) = n^2 - 14$.
   (c) Prove that there exist infinitely many positive integers n such that $s(n) = n^2 - 14$.

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